Piyush Hinduja

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Education

Master of Science in Computer Science GPA: **3.91**/4.0
University of Utah *2023 – 2025*

Bachelor of Engineering in Computer Engineering GPA: **3.85**/4.0
University of Mumbai *2019 – 2023*

Skills

Languages: *Expert (5+ yrs)*: Python, Java, JavaScript *Intermediate (3+ yrs)*: C, C++ *Beginner (1+ yrs)*: PHP, Kotlin, SQL, Shell
AI/ML: LLMs, PyTorch, HuggingFace, Langchain, TensorFlow, Scikit-Learn, BERT models, EasyOCR, spaCy
Data Processing: RESTful APIs, Pandas, PySpark, Databricks, OCRs, Matplotlib, Seaborn, DuckDB, PowerBI, Tableau
Applications and Databases: FastAPI, Streamlit, Bash, Preswald, MongoDB, MySQL, Amazon RDS, Firebase, ChromaDB
Cloud & Miscellaneous: AWS (S3, SageMaker, Texteract, IAM), Azure, Docker, Kubernetes, Git, CI/CD, Agile/SCRUM, JIRA

Work Experience

AI Engineer (formerly Data Analyst) March 2024 – Present
University of Utah *Salt Lake City, Utah*

- Building language models to analyze Earnings Conference Calls and 10-K, 10-Q, and 8-K filings, identifying correlations and executive statements linked to future performance trends
- Implemented advanced NLP analysis using **encoder-only model FinBERT, PyTorch, and Transformers** for financial document processing, **improving the prediction accuracy from 66% to 94.13%**
- Reduced per-filing **processing time by 75% using multi-cores and multi-thread environments** to parallelize file processing, API requests, and text cleaning operations
- Developed robust **prompt engineering** solutions to integrate LLMs & GenAIs (OpenAI, Anthropic, and Gemini) for automated financial text analysis, comparing AI-generated insights against traditional algorithms

Machine Learning Researcher August 2024 – April 2025
FLUX Research Group *Salt Lake City, Utah*

- Engineered automated deployment system using **Python and Linux Bash to streamline NVIDIA/CUDA toolkit** integration on high-performance computing clusters, reducing overhead by **90%**
- Analyzed wireless spectrum data to uncover scientific insights and **optimize power distribution across 20+ campus routing devices**

Full Stack Developer Intern June 2021 – July 2021
Exposys Data Labs *Bangalore, India*

- Spearheaded the launch of a tourism project, integrating key features to **enhance user experience** and deliver valuable information about 50+ Indian travel destinations
- Utilized **MERN stack** for an efficient frontend and backend integration, creating a responsive, user-friendly interface and enabling content management

Projects

SEC Comment Letter Analysis | *LLMs, OCR, Claude API*

- Engineered a rolling-window algorithm to process and bundle SEC comment letter conversations, analyzing data from **50,000+ companies** via the EDGAR database
- Designed an **OCR pipeline using the EasyOCR library** to extract text from PDFs with **95% accuracy**, processing 2-page documents in under 2 seconds
- Constructed prompt-based labeling system using **Claude API** to classify letter content with **98.25% accuracy**, integrating metadata for contextual inference

Road Damage Detection | *Object-Detection, YOLOv5, FasterRCNN* GitHub Link

- Collaborated with a team of four to develop a road damage detection system using **YOLOv5 and FasterRCNN**, integrating a user-friendly web interface (deployed via **Streamlit**), leveraging the RDD-2020 dataset
- Achieved model accuracies of **52.67% with YOLOv5 and 44.45% with FasterRCNN** by training and fine-tuning on different hyper-parameters
- The system plays a vital role, delivering a **70% improvement in road damage detection accuracy** over manual methods, helping lower the frequency of accidents and fatalities

Dependency Parser | *PyTorch, GloVe Embeddings* GitHub Link

- Architected an end-to-end **Transition-Based Dependency Parser in PyTorch**, implementing a multi-class classifier for predicting parsing actions based on stack and buffer states
- Designed modular pipeline incorporating data preprocessing, model architecture, training workflow, and evaluation metrics
- Integrated **GloVe word embeddings** for enhanced semantic representation, contributing to **UAS of 0.76 and LAS of 0.705**

Language Modeling | *NLP, RNN, N-gram, PyTorch* GitHub Link

- Developed **character-level LSTM model** in PyTorch to predict character sequences based on **50-character context windows**
- Implemented teacher forcing during training to prevent error propagation, using ground truth characters instead of predictions for subsequent timesteps
- Achieved **perplexity of 9.43** through systematic model architecture optimization and training methodology